Faithful Bounded Representations of Topologically Residually Nilpotent Banach–Lie Algebras

Partial-result packet for arXiv:1108.5563

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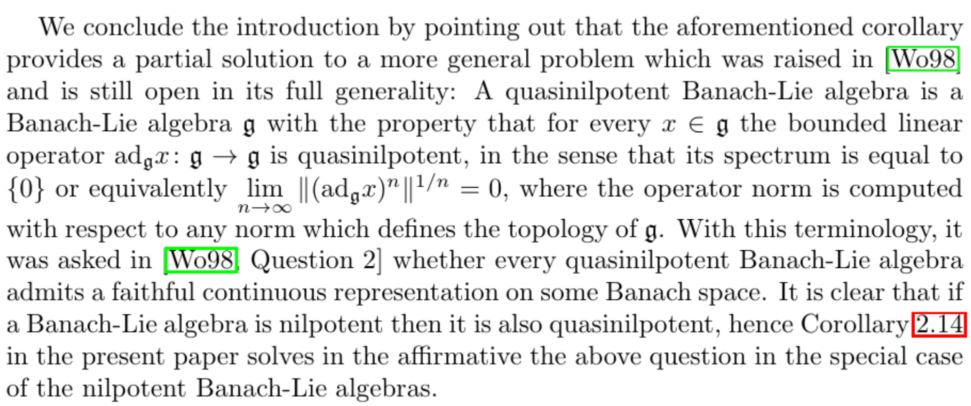
Status: candidate substantial partial result, likely valid; human review requested. Every Banach–Lie algebra separated by continuous homomorphisms into nilpotent Banach–Lie algebras has a faithful bounded representation on a Banach space. In particular, this holds when the intersection of the closed lower-central series is zero. This strictly extends the source's nilpotent theorem and gives a positive answer to Wojtyński's question for all topologically residually nilpotent quasinilpotent Banach–Lie algebras. The general quasinilpotent case remains open.

1 Source question

The source is Ingrid Beltiţă and Daniel Beltiţă, *Faithful representations of infinite-dimensional nilpotent Lie algebras*, arXiv:1108.5563 [1]. On page 2 it recalls Wojtyński's question:

> Does every quasinilpotent Banach–Lie algebra admit a faithful continuous representation on some Banach space?

Here quasinilpotent means that $\mathrm{Spec}(\mathrm{ad}\, x) = \{0\}$ for every $x \in \mathfrak{g}$. The source proves the affirmative answer when $\mathfrak{g}$ is nilpotent. The result below covers a broad inverse-limit-like class without any finite bound on the nilpotency classes of its quotients.



We conclude the introduction by pointing out that the aforementioned corollary provides a partial solution to a more general problem which was raised in [Wo98] and is still open in its full generality: A quasinilpotent Banach-Lie algebra is a Banach-Lie algebra $\mathfrak{g}$ with the property that for every $x \in \mathfrak{g}$ the bounded linear operator $\mathrm{ad}_{\mathfrak{g}} x \colon \mathfrak{g} \to \mathfrak{g}$ is quasinilpotent, in the sense that its spectrum is equal to $\{0\}$ or equivalently $\lim\limits_{n\to\infty} \|(\mathrm{ad}_{\mathfrak{g}} x)^n\|^{1/n} = 0$, where the operator norm is computed with respect to any norm which defines the topology of $\mathfrak{g}$. With this terminology, it was asked in [Wo98, Question 2] whether every quasinilpotent Banach-Lie algebra admits a faithful continuous representation on some Banach space. It is clear that if a Banach-Lie algebra is nilpotent then it is also quasinilpotent, hence Corollary 2.14 in the present paper solves in the affirmative the above question in the special case of the nilpotent Banach-Lie algebras.

Figure 1: The open quasinilpotent Banach–Lie representation question in the source PDF, page 2.

2 Proof intuition

There are two small ideas. First, the source represents an N-step nilpotent Banach–Lie algebra on polynomials of degree at most N. Every representing operator strictly lowers polynomial degree. Weighting the degree-m summand by a large factor R^m therefore makes the whole representation arbitrarily small in operator norm without changing its faithfulness or Lie relations.

Second, take all nilpotent quotients that separate the original algebra, make each quotient representation contractive, and place them on one ℓ_2-direct sum. The diagonal action is bounded because the coordinate norms are uniformly bounded; it is faithful because some quotient detects every nonzero element. The renorming step is what prevents the norms of the nilpotent representations from diverging with their nilpotency classes.

3 Definitions and statement

Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. A representation of a Banach–Lie algebra $\mathfrak{g}$ is a bounded linear Lie homomorphism $\rho : \mathfrak{g} \to \mathcal{B}(E)$ for a Banach space E over $\mathbb{K}$. We call $\mathfrak{g}$ *Banach-residually nilpotent* if there is a family of continuous Lie homomorphisms $q_i : \mathfrak{g} \to \mathfrak{h}_i$ into nilpotent Banach–Lie algebras such that $\bigcap_i \ker q_i = \{0\}$.

Define the closed lower-central series by

$$\mathcal{C}^1\mathfrak{g} = \mathfrak{g}, \qquad \mathcal{C}^{n+1}\mathfrak{g} = \overline{[\mathfrak{g}, \mathcal{C}^n\mathfrak{g}]}.$$

We say that $\mathfrak{g}$ is *topologically residually nilpotent* if $\bigcap_{n\geq 1} \mathcal{C}^n\mathfrak{g} = \{0\}$.

Theorem 1 (Residual nilpotence implies linearity)**.** *Every Banach-residually nilpotent Banach–Lie algebra admits a faithful bounded representation on a Banach space. More precisely, the representation can be chosen contractive.*

Corollary 2. *Every topologically residually nilpotent Banach–Lie algebra has a faithful bounded representation. Consequently Wojtyński's question has an affirmative answer for every quasinilpotent Banach–Lie algebra whose closed lower-central series has zero intersection.*

4 Proof

We first extract a quantitative form of the source's nilpotent construction.

Lemma 3 (Arbitrarily small nilpotent representations)**.** *Let $\mathfrak{h}$ be a nilpotent Banach–Lie algebra. For every $\varepsilon > 0$ there are a Banach space E and a faithful bounded representation $\pi : \mathfrak{h} \to \mathcal{B}(E)$ such that*

$$\|\pi(x)\| \leq \varepsilon\|x\| \qquad (x \in \mathfrak{h}).$$

Proof. Suppose $\mathfrak{h}$ has nilpotency class at most N. Write

$$\mathcal{P}_N(\mathfrak{h}, \mathbb{K}) = \bigoplus_{m=0}^{N} \mathcal{P}^m(\mathfrak{h}, \mathbb{K})$$

for the Banach space of continuous scalar polynomials of degree at most N, decomposed into homogeneous degrees. The differentiated left-regular construction in [1, Lemmas 2.11–2.12 and Corollary 2.14] gives a bounded Lie homomorphism

$$d\lambda : \mathfrak{h} \longrightarrow \mathcal{B}(\mathcal{P}_N(\mathfrak{h}, \mathbb{K}))$$

such that

$$d\lambda(x)\mathcal{P}^m(\mathfrak{h},\mathbb{K}) \subseteq \bigoplus_{k=0}^{m-1} \mathcal{P}^k(\mathfrak{h},\mathbb{K}). \tag{1}$$

It is faithful already on the full polynomial space: for $\xi \in \mathfrak{h}^* \subset \mathcal{P}_N$,

$$(d\lambda(x)\xi)(0) = \frac{d}{dt}\bigg|_{t=0} \xi((-tx) * 0) = -\xi(x),$$

so $d\lambda(x) = 0$ implies $x = 0$ by Hahn–Banach.

Choose fixed Banach norms $\|\cdot\|_m$ on the homogeneous summands. Since there are only finitely many blocks in (1), there is a finite constant C_N such that, writing P_k for homogeneous projection,

$$\sum_{0\le k<m\le N} \|P_k d\lambda(x)\phi_m\|_k \le C_N\|x\| \sum_{m=0}^{N} \|\phi_m\|_m.$$

For $R \ge 1$, give the same vector space the equivalent norm

$$\|\phi\|_R = \sum_{m=0}^{N} R^m \|\phi_m\|_m.$$

Every nonzero block drops degree by at least one, hence

$$\|d\lambda(x)\phi\|_R \le \frac{C_N}{R}\|x\|\|\phi\|_R.$$

Taking $R \ge C_N/\varepsilon$ proves the lemma. The underlying linear map and Lie identities are unchanged by this equivalent renorming. □

Proof of Theorem 1. Let $(q_i : \mathfrak{g} \to \mathfrak{h}_i)_{i\in I}$ be a separating family. By Lemma 3, choose a faithful representation $\pi_i : \mathfrak{h}_i \to \mathcal{B}(E_i)$ so small that

$$\|\pi_i\|\,\|q_i\| \le 1$$

(with no restriction needed when $q_i = 0$). Let

$$E = \left(\bigoplus_{i\in I} E_i\right)_{\ell_2}$$

and define the diagonal operator

$$\rho(x)(v_i)_{i\in I} = \big(\pi_i(q_i x)v_i\big)_{i\in I}.$$

Then

$$\|\rho(x)\| = \sup_i \|\pi_i(q_i x)\| \le \|x\|,$$

so $\rho : \mathfrak{g} \to \mathcal{B}(E)$ is bounded. It is a Lie homomorphism coordinatewise. If $\rho(x) = 0$, faithfulness of every π_i gives $q_i(x) = 0$ for all i, and the separating property gives $x = 0$. Thus ρ is faithful. □

Proof of Corollary 2. For $n \ge 1$, the quotient

$$q_n : \mathfrak{g} \longrightarrow \mathfrak{g}/\mathcal{C}^{n+1}\mathfrak{g}$$

is a continuous homomorphism into a nilpotent Banach–Lie algebra of class at most n. The kernels intersect in $\bigcap_n \mathcal{C}^{n+1}\mathfrak{g} = \{0\}$, so Theorem 1 applies. □

5 A strict nonnilpotent quasinilpotent example

Example 4. *Let $E = \ell_2(\mathbb{N}_0)$ with basis (e_k) and define the weighted shift*

$$De_k = 2^{-k} e_{k+1}.$$

Then

$$\|D^n\| = 2^{-n(n-1)/2}, \qquad \|D^n\|^{1/n} \longrightarrow 0,$$

so D is quasinilpotent but not nilpotent. Form

$$\mathfrak{g}_D = E \rtimes_D \mathbb{K}, \qquad [(x,a),(y,b)] = (aDy - bDx, 0).$$

Relative to $E \oplus \mathbb{K}$, every adjoint operator has block form

$$\mathrm{ad}_{(x,a)} = \begin{pmatrix} aD & -Dx \\ 0 & 0 \end{pmatrix},$$

and therefore has spectrum $\{0\}$. Thus $\mathfrak{g}_D$ is quasinilpotent. It is not nilpotent because $D^n \neq 0$ for every n. On the other hand,

$$\mathcal{C}^{n+1}\mathfrak{g}_D = \overline{D^n E} \oplus \{0\} = \overline{\mathrm{span}}\{e_n, e_{n+1}, \ldots\} \oplus \{0\},$$

whose intersection is zero. Corollary 2 therefore applies.

6 Scope, verification, and novelty

The result does not prove that every quasinilpotent Banach–Lie algebra is topologically residually nilpotent. Individual quasinilpotence of each $\mathrm{ad}\, x$ supplies no evident uniform joint-spectral-radius control on mixed commutators, and the closed lower-central residual may be nonzero. Nor does the diagonal construction separate an element lying in every nilpotent quotient. Those are the precise remaining obstructions to upgrading this packet to a full answer.

Verification notes. There is no computational dependency. Review should focus on three points: (i) the source's degree-lowering inclusion (1) on the full polynomial space, (ii) the factor R^{-1} in the weighted norm, and (iii) the uniform boundedness of the diagonal ℓ_2 action for an arbitrary index set. The example's spectral assertion follows from the spectrum of a triangular 2×2 operator matrix.

Novelty check. On 11 August 2026, the four run indexes were searched for the arXiv id, "quasinilpotent Banach–Lie", "residually nilpotent Banach–Lie", and "faithful representation". Bounded primary-source web searches used the exact phrases "residually nilpotent Banach–Lie algebra faithful representation", "pro-nilpotent Banach Lie linear", and the source and Wojtyński titles. We also checked the 2026 arXiv version of Neeb's infinite-dimensional Lie theory text, which records the nilpotent polynomial representation theorem and Wojtyński's BCH theorem separately. No source located the weighted-renorming plus diagonal-quotient theorem above. Because the argument is short and may be folklore, novelty confidence is moderate rather than high.

Human-review recommendation. Review as a substantial partial result. The proof is elementary once the source's polynomial representation is imported, but it expands the known positive class from a single nilpotency class to arbitrary separating towers of nilpotent Banach quotients.

References

[1] I. Beltiţă and D. Beltiţă, *Faithful representations of infinite-dimensional nilpotent Lie algebras*, Forum Math. **27** (2015), 255–267; arXiv:1108.5563.

[2] W. Wojtyński, *Quasinilpotent Banach–Lie algebras are Baker–Campbell–Hausdorff*, J. Funct. Anal. **153** (1998), 405–413.

[3] K.-H. Neeb, *Infinite-Dimensional Lie Groups*, arXiv:2602.12362 (2026), sections on linear and quasinilpotent Banach–Lie algebras.